

Use the theory of evolution by natural selection to explain past and present diversity and analyse the scientific evidence supporting the theory

Learning intention

- use the theory of evolution by natural selection to explain past and present diversity and analyse the scientific evidence supporting the theory.

Teach and model

- examining biodiversity as a function of evolution.
- relating genetic characteristics to survival and reproductive rates.
- outlining processes involved in natural selection including variation, isolation and selection.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.